

K2cloud Space State Transition and Status

Technical Note

Date: 16 April 2025

Version: 1.0

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1. Overview

This technical note outlines the lifecycle states and corresponding status indicators for K2Cloud Spaces, which represent tenant-specific runtime environments deployed and managed on Kubernetes. These states reflect both the intended operational commands (such as creation, deployment, and termination) and the outcomes of those operations (such as failure or success), as reported through the K2cloud Orchestrator control plane.

The document explains how K2view's distributed system components — including K2Cloud Agent, Kubernetes, and K2cloud Orchestrator — collaborate to derive and track the status of a space. It clarifies how state information is stored, how the system responds to environmental conditions (e.g., deployment errors or runtime health signals), and how this data propagates through the user interface.

2. System Roles in Determining Space State and Status

2.1. K2Cloud Agent

Role: The K2Cloud Agent is a key execution component deployed in the Kubernetes cluster. It is responsible for enacting the directives of the K2cloud orchestration service and monitoring space-level operations, acting as a bridge between the K2cloud Orchestrator and the Kubernetes environment.

Responsibilities:

- **Polling and Monitoring:** Regularly polls Kubernetes to assess the health and status of resources such as pods, services, deployments, and jobs associated with a Fabric space.
- **Execution:** Executes control operations (e.g., deploy, start, pause, terminate) as K2cloud Orchestrator requests.
- **Status Reporting:** Aggregates runtime data (e.g., pod conditions, job completion status, errors) and reports it to K2cloud Orchestrator via the Mailbox API.
- **Error Propagation:** Detects Kubernetes-level issues (e.g., image pull errors, failed deployments) and converts them into higher-level status codes such as Failed or Error.

2.2. Kubernetes

Role: Kubernetes provides the runtime layer for deploying and managing containerized Fabric spaces.

Responsibilities:

- **State Management:** Tracks the lifecycle and state of pods, deployments, config maps, and other resources.
- **Health Signals:** Emits events and status conditions (e.g., Running, Failed, Pending) used by the agent to determine the real-time state of Fabric components.

- **Failure Handling:**
 - Pod-level errors (e.g., readiness probe failures) can affect deployment success.
 - Resource constraints or quota issues can stall or crash spaces, triggering a Failed or Error state.

The agent interprets these Kubernetes events and translates them into Fabric-aware statuses.

2.3. K2cloud Orchestrator

Role: K2cloud Orchestrator is the centralized control plane for managing spaces and services across environments. It serves as the UI, configuration system, and state registry.

Responsibilities:

- **Initiation:** Sends commands to the agent (e.g., create space, deploy code) as JSON payload via the Mailbox REST service.
- **Mailbox REST Service Communication:** Uses a mailbox-based messaging pattern to asynchronously send and receive execution results from the agent via a callback.
- **Status Aggregation:** Maintains a record of each space's current status.
- **State Presentation:** Drives the UI status (e.g., color indicators) based on the latest agent reports.
- **Recovery and Refresh:** Supports refresh and remediation actions that allow users to reset or update a stale or failed state.

3. State Transition and Status Summary

This section summarizes the **state transitions** and corresponding **color-coded statuses** for K2Cloud Spaces, as maintained by the K2cloud service and reflected in the user interface (UI).

Color Semantics

- **Grey:** Denotes transitional or inactive state.
- **Orange:** Failure during non-critical operations.
- **Red:** Severe issue requiring investigation.
- **Blue:** Indicates the system is healthy and running.

State table

State	Color	Description
Creating	Grey	Initial phase of provisioning a space.
Starting	Grey	Space is initializing post-creation or post-pause.
Restarting	Grey	Triggered when the space is being restarted.
Paused	Grey	Space is idle and does not consume resources actively.
Pausing	Grey	Transitioning from Running to Paused state.
Resuming	Grey	Transitioning from Paused to Running state.
Terminating	Grey	Space is being shut down or deleted.
Terminated	Grey	Space has been shut down or deleted.
Waiting	Grey	Transient state during initial space setup, before creation begins.
Deploying	Grey	Deployment in progress. This may lead to Running or Failed.
Failed	Orange	Operation failed — typically deployment or creation-related.
Error	Red	Critical issue or unrecoverable error occurred.
Running	Blue	Space is operational and running as expected.

4. Key Behaviors and Insights

4.1. Status Persistence

- Each space has a status column in the **space table**, which reflects its current state.
- If a deployment or operation fails, the failure state is persisted by the K2cloud service and remains until an explicit update occurs (e.g., a successful deployment or status refresh).

4.2. Polling and API Integration

- The K2Cloud Agent regularly polls for space activity.
- The **API response** includes the space's status, which is rendered in the GUI using the associated color.
- The GUI updates the visual status based on this API response.

4.3. Transition Behavior

- A space typically transitions through the following lifecycle:
 - Waiting → Creating → Starting → Running
- The Deploying state is tied explicitly to code deployment and leads to:
 - Running (success)
 - Failed (failure)
- Once a state is set (e.g., Failed), it is not automatically cleared and must be updated through:
 - Deploy action
 - Refresh Status
 - Other lifecycle actions (e.g., Resume)

4.4. Deployment-Specific States

Deployment status is nuanced and deserves special mention:

- **Deploying:** Indicates a deployment operation is in progress.
- **Failed:** Deployment was attempted but failed (e.g., compilation, runtime error).
- **Running:** Deployment was successful, and the space is operational.
- **Error:** Severe issues or environment misconfiguration.